

Understanding Arabic Translation Challenges in Multilingual Models: A Data-Centric and Model-Level Analysis

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Abstract

We present an empirical study of Arabic-centric multilingual machine translation across six translation directions involving Arabic, English, Hindi, and Urdu. Our analysis identifies Arabic as the strongest distributional outlier, with the largest vocabulary, highest lexical diversity, lowest higher-order n-gram overlap, and greatest train–evaluation divergence, indicating higher translation difficulty and severe vocabulary sparsity. To assess the impact of these properties, we evaluate three state-of-the-art multilingual translation models, NLLB-200, MADLAD-400, and GemmaX2, under a unified fine-tuning framework. Fine-tuning consistently improves performance across all language pairs, increasing average BLEU from 14.13 to 26.19, COMET from 0.7338 to 0.8306, and ChrF2++ by 16.7 points while reducing TER by 18.5 points. MADLAD-400 achieves the best overall performance, while GemmaX2 shows the largest adaptation gain, improving from 4.26 to 25.73 BLEU after supervised fine-tuning. Human evaluation confirms substantial gains in adequacy and fluency. Arabic-source translation remains the most challenging direction despite significant adaptation gains, demonstrating that translation robustness depends not only on model capacity but also on corpus characteristics and train–evaluation distribution mismatch. We advocate Arabic-aware evaluation that combines corpus-level diagnostics with benchmark metrics to better understand and improve multilingual machine translation systems.

1 Introduction

Cross-regional communication between the Arab world and South Asia increasingly relies on reliable machine translation (MT), as trade, migration, digital media, and educational exchange intensify (Astute Analytica, 2024; GCC Secretariat General, 2024). Recent multilingual foundation models have advanced translation quality by exploiting large-

scale pretraining and cross-lingual transfer, making MT central to scalable cross-lingual access and interaction (Costa-jussá et al., 2022; Kudugunta et al., 2023).

However, progress remains uneven for morphologically rich and comparatively low-resource languages. Arabic is especially challenging because of its complex morphology, lexical variation, and orthographic diversity, while translation with Hindi and Urdu adds typological distance in script, syntax, and structure. Existing benchmarks still underrepresent Arabic-centric translation involving South Asian languages (Sajjad et al., 2020).

This study asks whether Arabic-centric directions are intrinsically harder than Hindi- and Urdu-related directions, how corpus properties such as vocabulary diversity, out-of-vocabulary rates, and train–evaluation distribution divergence shape translation quality, and how multilingual architectures respond to lexical and structural mismatch.

We benchmark NLLB-200 (Costa-jussá et al., 2022), MADLAD-400 (Kudugunta et al., 2023), and GemmaX2 (Cui et al., 2025) across six bidirectional Arabic-centric directions involving Arabic, English, Hindi, and Urdu. Using zero-shot and supervised fine-tuned settings, corpus diagnostics, and human evaluation, we show that Arabic is a distributional outlier, fine-tuning improves all directions, and Arabic-source translation remains the most difficult.

2 Related Work

Recent advancements in machine translation utilize massively multilingual foundation models like NLLB-200 and MADLAD-400 to facilitate cross-lingual transfer, though supervised fine-tuning remains critical to overcome domain mismatches, particularly for generative models like TranslateGemma (Rhaffouli and Boughaba, 2026). A persistent challenge in Arabic-centric translation to

South Asian languages is severe vocabulary sparsity driven by rich non-concatenative root-pattern morphology. Historically, addressing this required complex latent variable models (Ataman et al., 2020). While modern subword tokenizers risk distorting these structures, recent studies show that standard BPE tokenization fragments Arabic words into substantially more subword units than Latin-script languages, reducing token efficiency and increasing sequence length, which negatively impacts multilingual model performance (Alakeel et al., 2026) (Hatch and Richardson, 2025). Nonetheless, the fundamental structural divergence between Arabic and typologically distant languages like Hindi or Urdu remains a major barrier. This disconnect is exacerbated by train-evaluation distribution divergence. Standard metrics like BLEU penalize valid morphological variations, masking true semantic adequacy in Arabic translation (Beseiso et al., 2022). Character-level metrics such as ChrF++ and its variants address this by computing character and word n -gram F-scores that are robust to inflectional variation without requiring neural reference models (Al-Khalifa et al., 2024), making them particularly appropriate for the morphologically rich languages evaluated in this study. Both traditional models and LLMs retain distinct “translationese” features compared to authentic human text (Kong and Macken, 2025). Consequently, improving Arabic MT requires shifting from model-centric metrics toward diagnostic, corpus-aware evaluation paradigms that account for structural mismatches and vocabulary coverage.

3 Data and Corpus Analysis

A comprehensive analysis of the bidirectional Arabic-English (En↔Ar), Arabic-Hindi (Hi↔Ar), and Arabic-Urdu (Ur↔Ar) parallel datasets was conducted to assess domain representation, structural integrity, and cross-split representativeness.

3.1 Dataset Domain and Thematic Structure

The parallel corpora consist of short multilingual text segments drawn from multiple domains, with the majority of samples originating from contemporary news and analytical reporting spanning 2023 to projected developments in 2026. The dataset is highly segmented and attribution-rich, frequently referencing news agencies, official statements, and institutional reports. Rather than forming continuous narratives, most samples function as self-

Language Pair	Train	Dev	Test
Ar-En	21,000	2,000	2,000
Ar-Hi	20,100	2,000	2,000
Ar-Ur	20,300	2,000	2,000

Table 1: Corpus scale and alignment statistics for the Arabic-centric bidirectional parallel datasets. All language pairs exhibit strict source–target alignment with comparable train, development, and test splits (1:10 ratio), enabling balanced evaluation across six translation directions: Ar→En, En→Ar, Ar→Hi, Hi→Ar, Ar→Ur, Ur→Ar.

contained informational units.

The major thematic domains represented in the corpus include:

- **Geopolitics & International Relations:** Covering regional diplomacy, strategic alliances, conflict developments, and Middle East peace initiatives.
- **Global Economy & Energy:** Including trade agreements, macroeconomic reforms, energy partnerships, and financial market developments.
- **Technology, Health & Public Affairs:** Featuring AI innovations, healthcare updates, digital infrastructure, and public-sector policy developments.

The corpora maintain strict sentence-level alignment across source and target languages, enabling reliable multilingual translation modeling across all evaluated language pairs.

3.2 Corpus Scale and Alignment Integrity

Table 1 summarizes the corpus statistics for all language pairs. The bidirectional parallel datasets are comparable in scale, enabling balanced evaluation across all six translation directions. While the non-Arabic sides (English, Hindi, and Urdu) contain negligible within-file duplication, the aggregated Arabic corpus exhibits substantial duplication due to repeated alignment across multiple language pairs. This repeated reuse of Arabic source text should be considered when interpreting training dynamics, as it reduces the effective diversity of independent Arabic training evidence.

3.3 Lexical Diversity and Distributional Analysis

Corpus-level analysis also reveals a pronounced Arabic-specific distributional shift, making Ara-

Language	Sent. Pairs	Mean Len	Unique Tok.	TTR	OOV (%)	Dup (%)
Arabic	61,400	26.58	98,062	0.1757	10.9	0.069
English	21,000	23.45	45,234	0.0312	4.6	0.030
Hindi	20,100	38.33	58,233	0.0872	2.0	0.040
Urdu	20,300	41.42	32,960	0.0392	2.1	0.030

Table 2: Corpus-level lexical statistics highlighting Arabic as the strongest distributional outlier. Arabic exhibits 2.17× higher unique token vocabulary than English, elevated Out-of-Vocabulary (OOV) rates (10.9%), and lower Type-Token Ratio (TTR=0.1757). Hindi and Urdu show comparable, low OOV rates (2.0%), indicating their higher vocabulary coverage relative to training corpus size. TTR = Type-Token Ratio; OOV = Out-of-Vocabulary rate.

bic the most challenging language in our benchmark (Ataman et al., 2020; Alakeel et al., 2026). Although all sentence-length distributions are positively skewed, Arabic remains the most compact, with an average length of 26.7 tokens per sentence, compared to 38.33 for Hindi and 41.42 for Urdu under whitespace tokenization. Despite its shorter sequences, Arabic exhibits the highest lexical diversity and strongest sparse-tail behavior. In the evaluation splits, Arabic reaches an out-of-vocabulary (OOV) rate of nearly 11%, indicating substantial vocabulary sparsity, whereas Hindi and Urdu maintain robust lexical coverage with OOV rates near 2%. This elevated Arabic OOV burden is primarily driven by named entities, numeric variants, transliterations, and rare surface forms.

To assess generalization difficulty, we further analyze train–evaluation n-gram overlap. While Hindi and Urdu preserve strong unigram and bigram overlap due to recurring syntactic patterns, Arabic overlap decays sharply, with trigram overlap dropping to approximately 10%. Exact-match analysis confirms zero cross-split sentence leakage across all language pairs, ensuring evaluation reflects genuine generalization rather than memorization.

Note that corpus-level lexical statistics reported in this section are computed using whitespace tokenization for language-agnostic comparison and should not be interpreted as the token sequences used by the evaluated models during training or inference.

4 Experimental Methodology

4.1 Experimental Design

To investigate the impact of corpus-level distribution shift on Arabic-centric multilingual machine translation (MT), we design a controlled benchmarking framework spanning six bidirectional translation directions involving Arabic, English, Hindi, and Urdu ($\text{En} \leftrightarrow \text{Ar}$, $\text{Hi} \leftrightarrow \text{Ar}$, $\text{Ur} \leftrightarrow \text{Ar}$).

Our experimental setup evaluates translation performance under both zero-shot and supervised fine-tuning settings, enabling analysis of inherent multilingual transfer capability as well as task-specific adaptation.

This design allows us to address three core research questions: (1) how linguistic divergence between Arabic and typologically distinct languages such as English, Hindi, and Urdu influences translation difficulty, (2) how corpus-level properties such as lexical diversity, out-of-vocabulary (OOV) rates, and train–evaluation divergence influence translation quality, and (3) Objective is to analyze how widely adopted multilingual foundation models handle Arabic-specific linguistic challenges under cross-lingual transfer adapt to these challenges under supervised training.

4.2 Model Selection

We evaluate three state-of-the-art multilingual translation systems representing distinct architectural paradigms. While these systems provide strong multilingual baselines, recent Arabic-native large language models such as Fanar Prime and ACEGPT have demonstrated superior performance on Arabic-centric benchmarks, highlighting the growing importance of language-specialized architectures for Arabic NLP (Team, 2025; AlKhamissi et al., 2025). We therefore interpret the evaluated multilingual systems as general-purpose baselines rather than upper-bound performance for Arabic translation.

- **NLLB-200 (3.3B):** A dense encoder–decoder multilingual neural machine translation model optimized for direct many-to-many translation without relying on English as a pivot.
- **MADLAD-400 (10B):** A large-scale T5-based multilingual sequence-to-sequence model trained on extensive multilingual corpora with strong cross-lingual transfer capa-

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bility.

- **GemmaX2 (~10B):** A decoder-only large language model adapted for multilingual translation using instruction-style prompting and supervised fine-tuning. In addition to task-specific adaptation, its reported multilingual performance benefits substantially from continual pre-training under a Parallel-First Monolingual-Second (PFMS) data mixing strategy, designed to strengthen cross-lingual alignment and translation capability (??).

These models were selected to enable systematic comparison between specialized translation systems, multilingual encoder–decoder architectures, and general-purpose generative language models.

4.3 Preprocessing and Tokenization

To preserve original lexical and orthographic characteristics — particularly the Perso-Arabic script shared between Arabic and Urdu — no heuristic normalisation was applied prior to tokenisation. Post-decoding, outputs were normalised to NFC to canonicalise Unicode codepoint variants within the shared script space, partially capturing the orthographic overlap that phonetic-space projection methods (Kumar et al., 2023) exploit more aggressively for Indic language pairs.

4.4 Training and Adaptation Protocol

Each model was evaluated under both zero-shot and supervised fine-tuned settings across all six translation directions. Fine-tuning was performed independently for each direction using a unified optimization pipeline.

Due to substantial differences in model scale and architecture, adaptation strategies varied across systems. NLLB underwent full-parameter fine-tuning, allowing complete task-specific adaptation of the model weights. In contrast, MADLAD-400 and GemmaX2 were adapted using parameter-efficient fine-tuning methods based on Low-Rank Adaptation (LoRA) and Quantized Low-Rank Adaptation (QLoRA), respectively. This design enabled efficient optimization of large-scale models while maintaining comparable training conditions across architectures.

4.5 Inference and Evaluation Protocol

To ensure fair comparison, all systems were decoded using deterministic beam-search, repetition

control, and length penalties to reduce hallucinations and excessive output expansion.

Translation quality was assessed using four complementary automatic metrics: BLEU, COMET-22, ChrF2++, and Translation Edit Rate (TER). While BLEU measures n -gram precision, it penalises valid morphological variants and is known to underperform for morphologically rich target languages such as Arabic (Beseiso et al., 2022). ChrF2++ computes character- and word-level n -gram F-scores and is widely adopted as the standard lightweight metric for such languages precisely because it captures inflectional variation without the high-resource dependency of neural metrics (Al-Khalifa et al., 2024; Popović, 2017). COMET-22 provides neural semantic adequacy scores, while TER measures the post-editing effort required to correct system output. This four-metric suite enables cross-directional comparison that is robust to both Arabic’s morphological complexity and Urdu and Hindi’s agglutinative character-level behaviour. In addition to automatic evaluation, human assessment will be conducted to evaluate adequacy and fluency.

This unified evaluation framework enables systematic analysis of how corpus characteristics and architectural design jointly influence Arabic-centric multilingual translation performance under both zero-shot and supervised adaptation settings.

5 Results and Analysis

Table 3 reports all four metrics for every system and direction under the baseline (zero-shot) and fine-tuned settings. We organise the analysis around the aggregate effect of adaptation (Figure 1), its direction-level structure (Figure 2), and the failure modes the averages obscure (§??).

5.1 Fine-tuning Gains

Fine-tuning helps uniformly: across all eighteen system–direction pairs, COMET-22, ChrF2++ and BLEU rise and TER falls, with no regressions. On macro average BLEU improves from 14.13 to 26.19, COMET-22 from 0.7338 to 0.8306, ChrF2++ by 16.7 points, and TER drops by 18.5 points.

The magnitude, however, splits into two regimes (Figure 1). The dedicated encoder–decoder systems gain modestly — NLLB +5.5 BLEU, MADLAD +9.2 BLEU on macro average — as already-competent models specialise to the corpus domain;

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Direction	Baseline				Fine-tuned			
	COMET	ChrF2++	BLEU	TER↓	COMET	ChrF2++	BLEU	TER↓
<i>NLLB-200</i>								
Ar→En	0.829	47.5	21.9	66.7	0.842	53.9	29.2	60.3
En→Ar	0.864	45.8	15.8	82.6	0.876	49.0	21.1	71.8
Ar→Hi	0.709	46.1	29.5	67.9	0.735	50.9	35.2	63.9
Hi→Ar	0.847	42.8	13.9	84.3	0.859	46.0	17.8	75.6
Ar→Ur	0.781	43.6	25.1	70.6	0.800	48.6	32.2	65.2
Ur→Ar	0.854	42.5	14.4	81.5	0.865	45.8	17.9	75.7
Avg.	0.814	44.7	20.1	75.6	0.829	49.0	25.6	68.7
<i>MADLAD-400</i>								
Ar→En	0.830	46.5	20.8	67.6	0.848	56.5	32.3	57.5
En→Ar	0.867	46.0	15.7	81.7	0.883	51.1	24.3	67.3
Ar→Hi	0.704	42.2	25.6	70.0	0.742	51.7	36.4	63.3
Hi→Ar	0.855	44.7	14.6	83.6	0.869	48.7	21.6	70.5
Ar→Ur	0.752	35.8	16.9	75.2	0.792	46.3	28.1	68.0
Ur→Ar	0.863	44.1	14.5	84.1	0.876	47.9	20.8	71.9
Avg.	0.812	43.2	18.0	77.0	0.835	50.4	27.3	66.4
<i>GemmaX2</i>								
Ar→En	0.831	47.7	21.9	67.4	0.844	56.4	31.3	60.5
En→Ar	0.601	10.5	3.1	122.4	0.874	48.8	21.7	70.5
Ar→Hi	0.442	0.9	0.1	101.9	0.726	49.6	34.2	68.8
Hi→Ar	0.578	0.8	0.1	128.8	0.859	45.8	18.4	75.7
Ar→Ur	0.461	0.6	0.2	101.0	0.794	47.6	31.1	69.2
Ur→Ar	0.541	1.2	0.2	128.2	0.867	45.4	17.6	76.6
Avg.	0.576	10.3	4.3	108.3	0.827	48.9	25.7	70.2

Table 3: Per-direction results for all three systems under the baseline (zero-shot) and fine-tuned settings. COMET denotes COMET-22 and for TER lower is better. Avg. is the macro-average over the six directions. The baseline block for GemmaX2 records the language-identity collapse discussed in §??.

MADLAD, the largest, ends strongest, leading COMET-22 on five of six directions. GemmaX2 is different: its macro BLEU rises from 4.26 to 25.73 and COMET-22 from 0.5757 to 0.8272, the largest gain by far. and adaptation brings the model to within 0.005 COMET-22 of the specialised systems. Notably, the largest COMET-22 gains fall where the baseline is weakest, so adaptation compresses the spread across directions rather than lifting all equally.

5.2 Direction-Level Structure

After fine-tuning the systems converge to a narrow band, and the residual variation is organised by direction, not by model (Figure 2). Ar→Hi is the hardest direction for all three (COMET-22 0.735/0.742/0.726), each system’s own worst score; since three architectures under three adaptation regimes agree, this is a property of the language pair, which shares neither script nor substantial vocabulary. Ar→Ur is consistently easier (COMET-22 $\approx$ 0.79–0.80) despite Urdu being equally low-resource, plausibly because its Perso-Arabic script shares characters and borrowed vocabulary with Arabic.

A directional asymmetry also runs through the table: translating *out of* Arabic yields far higher BLEU than translating *into* it (32.2 vs. 20.1 on macro average), yet into-Arabic directions score *higher* on COMET-22 (0.870 vs. 0.791). Arabic’s rich morphology and free word order mean a single reference covers few of the acceptable surface forms, so *n*-gram metrics penalise valid into-Arabic output that COMET correctly rewards. BLEU should therefore not be read as a cross-direction ranking when Arabic is the target.

6 Limitations

The corpus is restricted to formal news and analytical reporting in MSA, standard Hindi, and formal Urdu. This limits generalization to informal, conversational, or social media-derived text, where dialectal Arabic variation, Roman-script code-switching in Hindi and Urdu, and non-standard orthography introduce lexical challenges substantially beyond those documented here (Sajjad et al., 2020; Mehmood et al., 2020). further exclusion of Arabic-native large language models such as Jais, ACEGPT, and Fanar, which increasingly serve as strong baselines for Arabic-centric

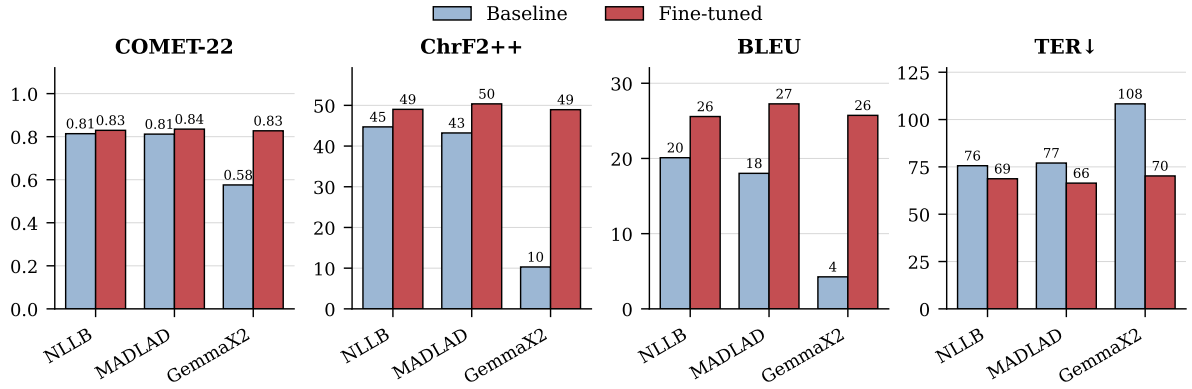


Figure 1: Macro-averaged quality across the six directions, baseline versus fine-tuned, per system. Adaptation improves every metric for every model; the gain is modest for the dedicated NMT systems and dramatic for GemmaX2, whose baseline averages are depressed by collapse on five of six directions (§??). Lower TER is better.

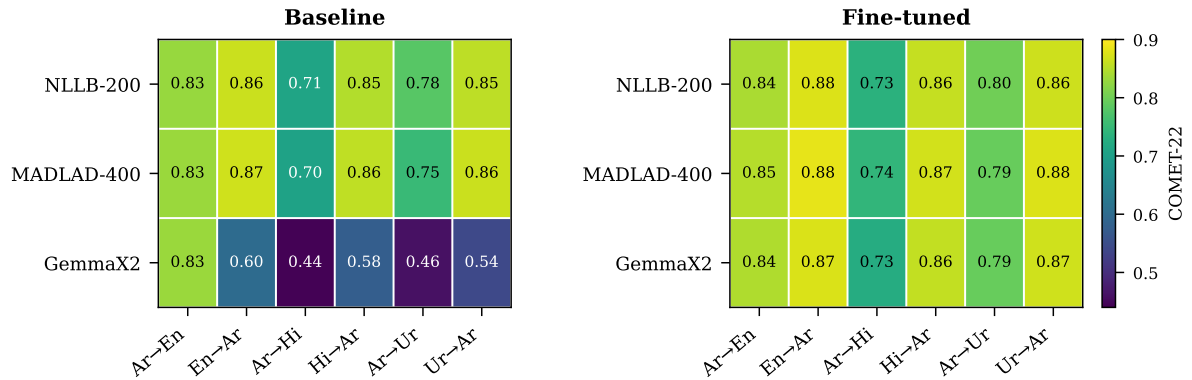


Figure 2: COMET-22 by system and direction, baseline (left) versus fine-tuned (right). The GemmaX2 collapse appears as a band confined to its row; after adaptation the three systems are nearly indistinguishable, with a residual hard column at Ar→Hi shared by all.

NLP. Their language-specific tokenization and pre-training may better address Arabic morphological challenges than general multilingual models. Future work should compare multilingual and Arabic-native architectures to quantify the trade-off between general multilingual capability and Arabic-specific optimization.

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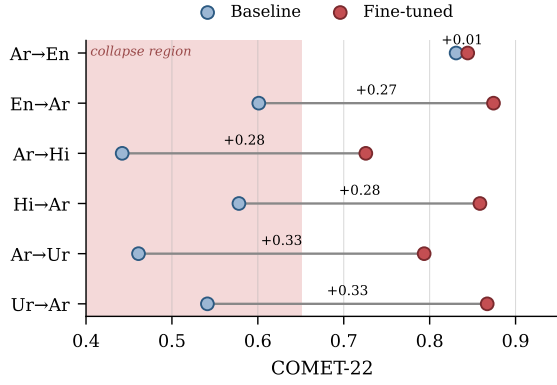


Figure 3: GemmaX2 COMET-22 before and after fine-tuning, per direction; labels give the gain. Every non-English target begins in the shaded collapse region and is recovered to the competitive band, while Ar→En barely moves.

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A Example Appendix

A.1 Fine-Tuning Methodology

Supervised fine-tuning was performed independently for each model and translation direction using a unified training pipeline with a fixed random seed (42), gradient checkpointing, linear learning-rate scheduling, and early stopping with a patience of two evaluation epochs.

Due to architectural and scale differences, adaptation strategies varied across models. **NLLB-200** underwent full-parameter fine-tuning, updating all 3.345B parameters in BF16 mixed precision without quantization. Memory efficiency was improved using the bitsandbytes 8-bit AdamW optimizer. Training used an effective batch size of 32, a peak learning rate of 5×10^{-5} , a warm-up ratio of 0.10, and weight decay of 0.01. Model selection was based on development-set COMET-22 performance.

For **MADLAD-400**, the base model weights were frozen and loaded in 8-bit precision, with adaptation performed using Low-Rank Adaptation (LoRA). LoRA modules were inserted into attention and feed-forward projections, yielding 94.4M trainable parameters ($\approx 0.88\%$ of total parameters). Fine-tuning used rank $r = 16$, scaling factor $\alpha = 32$, dropout of 0.05, a learning rate of 5×10^{-5} , and an effective batch size of 32.

Checkpoint selection was guided by development-set COMET-22 or validation loss, depending on the translation direction.

GemmaX2 was adapted using Quantized Low-Rank Adaptation (QLoRA), with frozen base weights loaded in 4-bit NF4 quantization and BF16 computation. LoRA layers were applied to all major linear projection modules, resulting in 54.0M trainable parameters. Training was limited to 3 epochs with an effective batch size of 32, a learning rate of 2×10^{-4} , warm-up ratio of 0.05, and weight decay of 0.01. Final checkpoints were selected based on minimum validation loss.